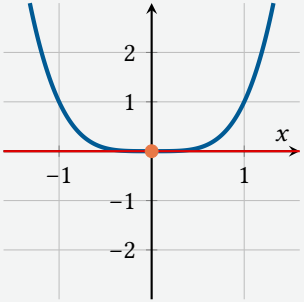
$g(x) = x^4$

$g''(0) = 0$

$g''(x) > 0$ on both sides

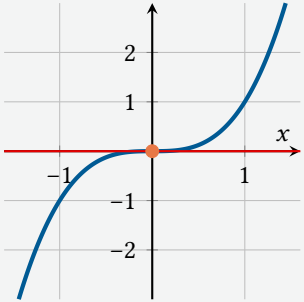


minimum, not an inflection point

$f(x) = x^3$

$f''(0) = 0$

$f''(x) < 0$ left, > 0 right



sign change confirmed: inflection point at $x = 0$